Points 6 Published

Details Questions

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Question 2.8 pts

Match each of the terms in the left column to the best-fitting phrase in the right column.

transgene

gene transferred by a scientist into an organism's genome
pronuclear injection

method of DNA transfer used for many vertebrates
floxed gene

useful for making conditional knockouts
AAV vector

genetically engineered viral genome that transfers a therapy gene
Cas9

endonuclease used for genome editing along with sgRNA
knockout mouse

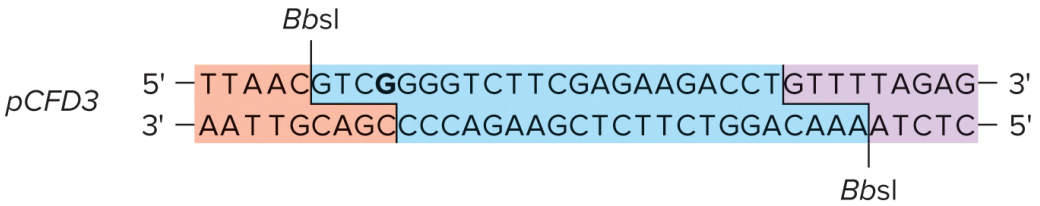
loss-of-function mutant through gene targeting
knockin mouse

contains additional or altered DNA through gene targeting
Cre recombinase

causes crossovers at loxP sites

Question

The following figure shows a part of a plasmid vector *pCFD3* for the expression of sgRNAs in mice. The orange sequences are part of a strong promoter (transcription from this promoter starts at the G in bold—which must be present—and goes from left to right). The purple sequences are a portion of the tracrRNA component of the sgRNA. After cutting the pCFD3 plasmid with the restriction enzyme BbsI (whose recognition site is also shown in the following figure), you will replace the blue sequences in the figure with sequences that will allow the expression of an sgRNA that targets a mouse gene called Nipp1.



*Bbs*I recognition site	5'	GAAGACNN^	3'
	3'	CTTCTGNNNNNN^	5'

The last part of the jigsaw puzzle you will need is the following sequence, which shows part of the Nipp1 gene including the triplet corresponding to the start codon. Capital letters are in the gene's first exon with the coding region in blue; lowercase letters are in the first intron. The Nipp1 protein is 383 amino acids long. Your assignment is to generate a knockout allele of this gene by inducing Cas9 to produce a double stranded break into the gene that will be repaired imprecisely by nonhomologous end-joining (NHEJ).

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MetThrAsnSerTyrAspIleHisSer
5'...GTTAAAAGTATGACTAACAGCTACGACATACACAGTTGgtgagtttggcatc... 3'
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Question 1.2 pts

List two possible 20nt crRNA sequences that can be designed to cut the Nipp1 gene. Indicate polarity. e.g, 5'-NNNNNNNNNNNNNNNNNNNN-3'. Remember that crRNA is an RNA molecule.

Sequence 1: [seq1]

Sequence 2: [seq2]

It doesn't matter which sequence you enter first. Give all nucleotides in uppercase.

Show Answers for seq1

5'-ACAGCUACGACAUACACAG

5'-ACAUACACAGUUGGUGAGU

Question 0.8 pts

Which of the two sequences above would be preferred to produce a null allele of the Nipp1 gene?

5'-ACAGCTACGACATACACAGT

Question

Discussion: Why is one sequence preferred over the other?

Question

Discussion: Diagram the pCFD3 vector after it has been cut with the BbsI enzyme. Don't worry about the small blue fragment that will be removed; the emphasis here is to show the 5' overhangs that will be made. The answer is at the very end of this quiz, but try not to look ahead.

Question 1.2 pts

Design two 24-nt DNA oligonucleotides that you could anneal together and clone directly into BbsI-cut pCFD3 vector so that the recombinant plasmid could express an sgRNA useful for making null mutations in the Nipp1 gene. Hint: there is no need to cut the oligonucleotide with BbsI.

Indicate polarity. e.g, 5'-NNNNNNNNNNNNNNNNNNNNNNNN-3'.

Sequence 1: [seq1]

Sequence 2: [seq2]

It doesn't matter which sequence you enter first. Give all nucleotides in uppercase.

Show Answers for seq1

5'-GTCGACAGCTACGACATACA

5'-AAACACTGTGTATGTCGTAG

Question

Discussion: Briefly outline what you would do with your recombinant plasmid to make a knockout mouse for the Nipp1 gene. The answer is at the very end of the quiz, but try not to look ahead.

Question

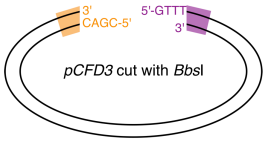
Discussion: Briefly outline how you would modify this technique to generate a knockin allele in which the first amino acid in the Nipp1 protein after the initiating Met (that is, Thr) would be changed to Ala. The answer is at the very end of the quiz, but try not to look ahead.

Question

Discussion: Briefly outline how you would modify this technique to make a conditional knockout mouse to only knockout the Nipp1 gene in eye cells. The answer is at the very end of the quiz, but try not to look ahead.

Question

Here is the answer to the question about diagramming the pCFD3 vector after cutting with the BbsI enzyme.



Question

Answer to how to make a knockout mouse for the Nipp1 gene: There are many possible ways. The following is one example.

i) Inject the plasmid into fertilized eggs. The injection mix also needs a source of Cas9 protein. You could either inject the Cas9 protein itself or a DNA in which the Cas9 coding region is placed downstream of a strong promoter. Note here, that you can use fertilized eggs directly instead of using mouse ES cells because of the higher efficiency of editing using the CRISPR/Cas9 system. ES cells are used in traditional knockout studies because homologous recombination alone is inefficent. ES cells allow researchers to grow cells in a dish and only select the small population of cells that are neomycin resistant and ganciclover resistant.

ii) These injected eggs are than placed into the oviduct of receptive females. Mice that were injected as embryos are born and crossed with wild-type (uninjected) mice as adults. Because NHEJ will produce a random indel mutation at the site, each individual could have a different mutation on both copies of the chromosome, one copy of the chromosome, or no mutation at all. By doing this cross with wild-type you are in effect isolating a single mutagenized chromosome.

iii) Some of the F1 progeny, with luck, will be heterozygous for a Nipp1 null mutation. You can identify lines in which this occurred using a PCR assay.

iv) Heterozygous F1 males and females can be crossed to each other to generate Nipp1 mutant homozygotes.

Question

Answer to how to make a knockin mouse for the Nipp1 gene: There are many possible ways. The following is one example.

Steps are very similar to generating the knockout mouse except you also inject linear double-stranded DNA molecules with the Thr codon (ACT) changed to an Ala codon (GCT). These double-stranded molecules would need to include sequences to the left and right of the codon in question. In some cells, repair of the double stranded break caused by Cas9 will occur by homologous recombination, resulting in replacement of the Thr codon with the Ala codon. For a single base pair change, the double-stranded DNA molecule that is introduced only needs to be 100-150bp long.

Question

Answer to how to make a conditional knockout mouse: There are many possible ways. The following is one example.

CRISPR/Cas9 could be used to flox a gene by knocking in a construct like the one in the figure that follows. All of the blue sequences are derived from the gene in which you want to make the conditional knockout. In the scenario of the Nipp1 gene, since you would target the first exon, the upstream region of homology would not be an Intron, but the exon sequence is the first exon.

Because this technique is so efficient, Neo^r is unnecessary. A sgRNA (to direct Cas9 to cleave in the region), Cas9, and the double-stranded DNA construct could be injected into fertilized eggs as transient transgenes.

You would go through the steps described above, to generate homozygous mice that have floxed Nipp1 gene. Finally, to make a conditional knockout in eye cells, you would cross these mice with other mice that only express Cre in the eye (how would you make these mice that only express Cre in the eye?). The progeny of this cross would knockout Nipp1, only in the eye.

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